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Case Study: Transparent & Explainable Residential Property Valuation

Applied Machine Learning, Leakage-Safe DAG Preprocessing, and Non-Causal Attribution in Ames, Iowa

Antigravity Applied Data Science

2026-10-07

1 Case Study Overview

1.1 Case Study Title

Transparent & Explainable Residential Property Valuation: Machine Learning Benchmarking, Leakage-Safe DAG Workflows, and Game-Theoretic Attribution in Ames, Iowa.

1.2 Problem Statement

Residential real estate property valuation represents one of the largest financial asset classes globally. In automated valuation modeling (AVM), real estate technology platforms often face an unacceptable compromise: 1. **Opaque Machine Learning Ensembles:** Complex tree ensembles (XGBoost, LightGBM) and deep neural networks minimize mean squared error, but operate as black boxes. When an appraisal drops by $40,000, lenders, underwriters, and property owners receive zero defensible explanation. 2. **Oversimplified Linear Hedonic Models:** Classical ordinary least squares (OLS) regression models provide transparent slope coefficients, but struggle with multicollinearity across dozens of attributes and cannot capture non-linear size-quality interactions, resulting in large prediction errors (exceeding $21,000 MAE).

Furthermore, black-box valuations in mortgage lending create severe regulatory compliance risks under the **Fair Housing Act** and the **Equal Credit Opportunity Act (ECOA)**, which mandate understandable explanations for adverse credit and valuation decisions.

1.3 Background & Project Context

This case study uses the canonical Ames Housing dataset (`mlr3data::ames_housing`) comprising 2,930 historical home sales in Ames, Iowa (2006–2010). Rather than training models on naive, leaky preprocessing scripts, we design an end-to-end machine learning system inside R using the modern `mlr3` ecosystem (`mlr3` , `mlr3pipelines` , `mlr3tuning`). All preprocessing transformations are contained within a Directed Acyclic Graph (DAG) to guarantee zero data leakage between training and validation folds.

1.4 Core Objectives

1. **Data Forensic Audit:** Systematically identify and treat structural dataset anomalies (zero-variance features, duplicate columns, and collinear scaling artifacts) prior to modeling.
2. **Leakage-Safe DAG Preprocessing & Feature Science:** Encapsulate all missing-value imputation, factor level collapsing, dummy encoding, and domain feature engineering inside an immutable `mlr3pipelines` DAG.
3. **Multi-Model Benchmark Tournament:** Benchmark 6 distinct regression model families across identical 5-fold cross-validation splits to establish performance baselines.
4. **Hyperparameter Optimization & Generalization Verification:** Tune the champion architecture using `paradox` parameter search and certify out-of-sample generalization on 586 untouched holdout properties.
5. **Dual Explainable AI (XAI) & Governance:** Implement global permutation feature importance and local **SHAP** (Shapley Additive Explanations) alongside **LIME** surrogates with explicit non-causal regulatory boundaries.

1.5 Target Prediction Variable

The objective is to predict **`Sale_Price`** (continuous target in USD) while decomposing each valuation into dollar-denominated positive enhancements and negative discounts relative to the Ames market baseline ($180,800).

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2 Dataset Provenance & Characteristics

2.1 Dataset Name & Source

- **Dataset Name:** Ames Housing Dataset
- **R Package Source:** `mlr3data::ames_housing`
- **Original Author:** Dean De Cock (2011), *Journal of Statistics Education*, Vol. 19, No. 3.
- **Geographic & Temporal Coverage:** Ames, Iowa, USA; sales recorded between 2006 and 2010.

2.2 Dimensions & Variable Types

The dataset contains **2,930 observations** (individual property transactions) and **81 variables**: * **1 Continuous Target:** `Sale_Price` * **47 Categorical Factors:** e.g., `Neighborhood`, `Bldg_Type`, `House_Style`, `Foundation`, `Exterior_1st` * **33 Integer Metrics:** e.g., `Gr_Liv_Area`, `Total_Bsmt_SF`, `Year_Built`, `Year_Remod_Add`, `Full_Bath`, `Garage_Cars` * **1 Numeric Metric:** `Lot_Area_m2` (collinear duplicate scale)

Hide

```
task_raw <- tsk("ames_housing")
dt_raw <- as.data.table(task_raw$data())

cat(sprintf("Observations : %s\n", format(nrow(dt_raw), big.mark = ",")))
```

```
## Observations : 2,930
```

Hide

```
cat(sprintf("Predictors   : %d (plus 1 target: Sale_Price)\n", ncol(dt_raw) - 1))
```

```
## Predictors   : 81 (plus 1 target: Sale_Price)
```

Hide

```
cat(sprintf("Target Range : %s to %s (Median: %s)\n",
            format_currency(min(dt_raw$Sale_Price)),
            format_currency(max(dt_raw$Sale_Price)),
            format_currency(median(dt_raw$Sale_Price))))
```

```
## Target Range : $12,789 to $755,000 (Median: $160,000)
```

2.3 Suitability for the Case Study

1. **Rich Structural Granularity:** Captures architectural layout, basement finish, garage size, and outdoor living space across entry-level starter homes and custom executive estates.
2. **Realistic Data Quality Challenges:** Contains missing values that denote structural absence (e.g., houses without garages or fireplaces) rather than missing data records, requiring nuanced imputation logic.
3. **High Dimensional Complexity:** 80+ features present realistic challenges regarding multicollinearity, high-cardinality factors, and non-linear interactions.

---

3 Data Preparation & Leakage-Safe Preprocessing

3.1 Forensic Audit of Intentional Dataset Anomalies

Prior to pipeline construction, a forensic audit confirmed three structural defects:

Hide

```
# 1. Audit Constant Feature: Misc_Feature_2
n_levels_misc2 <- uniqueN(dt_raw$Misc_Feature_2, na.rm = TRUE)
cat(sprintf("[Audit 1] Misc_Feature_2 non-NA distinct levels: %d -> Zero-variance co
        nstant.\n", n_levels_misc2))
```

```
## [Audit 1] Misc_Feature_2 non-NA distinct levels: 1 -> Zero-variance constant.
```

Hide

```
# 2. Audit Duplicate Feature: Condition_3 vs Condition_2
is_duplicate <- identical(dt_raw$Condition_2, dt_raw$Condition_3)
cat(sprintf("[Audit 2] Condition_3 identical to Condition_2? %s -> Exact duplicat
        e.\n", is_duplicate))
```

```
## [Audit 2] Condition_3 identical to Condition_2? TRUE -> Exact duplicate.
```

Hide

```
# 3. Audit Collinear Scale: Lot_Area vs Lot_Area_m2
r_collinear <- cor(dt_raw$Lot_Area, dt_raw$Lot_Area_m2, use = "complete.obs")
cat(sprintf("[Audit 3] Correlation(Lot_Area, Lot_Area_m2): %.4f -> Literal scale clo
        ne.\n", r_collinear))
```

```
## [Audit 3] Correlation(Lot_Area, Lot_Area_m2): 1.0000 -> Literal scale clone.
```

Interpretation: `Misc_Feature_2` is dropped because constant columns add zero predictive signal. `Condition_3` and `Lot_Area_m2` are pruned to eliminate collinear instability.

3.2 Domain-Driven Feature Engineering

Real estate appraisal fundamentally depends on synergistic interactions that raw linear models miss. Six domain features were engineered:

1. **`Qual_x_LivingArea` :** Multiplicative interaction between numerical quality score (1–10) and above-ground living area (\(r = 0.8454\) with `Sale_Price`). An extra 200 sq ft in a luxury home yields far greater dollar appreciation than in an entry-level home.
2. **`Total_Living_Area` :** Above-ground living area plus finished basement square footage (\(Gr_Liv_Area + \max(0, Total_Bsmt_SF - Bsmt_Unf_SF)\)).
3. **`Total_Bathrooms` :** \(Full_Bath + 0.5 \times Half_Bath + Bsmt_Full_Bath + 0.5 \times Bsmt_Half_Bath\).
4. **`Total_Porch_SF` :** Aggregates wood decks, open porches, enclosed porches, and screen rooms.
5. **`Age_At_Sale` & `Years_Since_Remodel` :** Disentangles chronological age from modern renovation investments.

Hide

```
# Feature engineering applied defensibly to data.table
dt_augmented <- add_engineered_features(dt_raw)
cor_qual_size <- cor(dt_augmented$Qual_x_LivingArea, dt_augmented$Sale_Price, use =
         "complete.obs")
cor_raw_size <- cor(dt_augmented$Gr_Liv_Area, dt_augmented$Sale_Price, use = "comple
         te.obs")

cat(sprintf("Correlation with Sale Price:\n"))
```

```
## Correlation with Sale Price:
```

Hide

```
cat(sprintf("  - Raw Above-Ground Living Area (Gr_Liv_Area) : r = %.4f\n", cor_raw_s
         ize))
```

```
##   - Raw Above-Ground Living Area (Gr_Liv_Area) : r = 0.7068
```

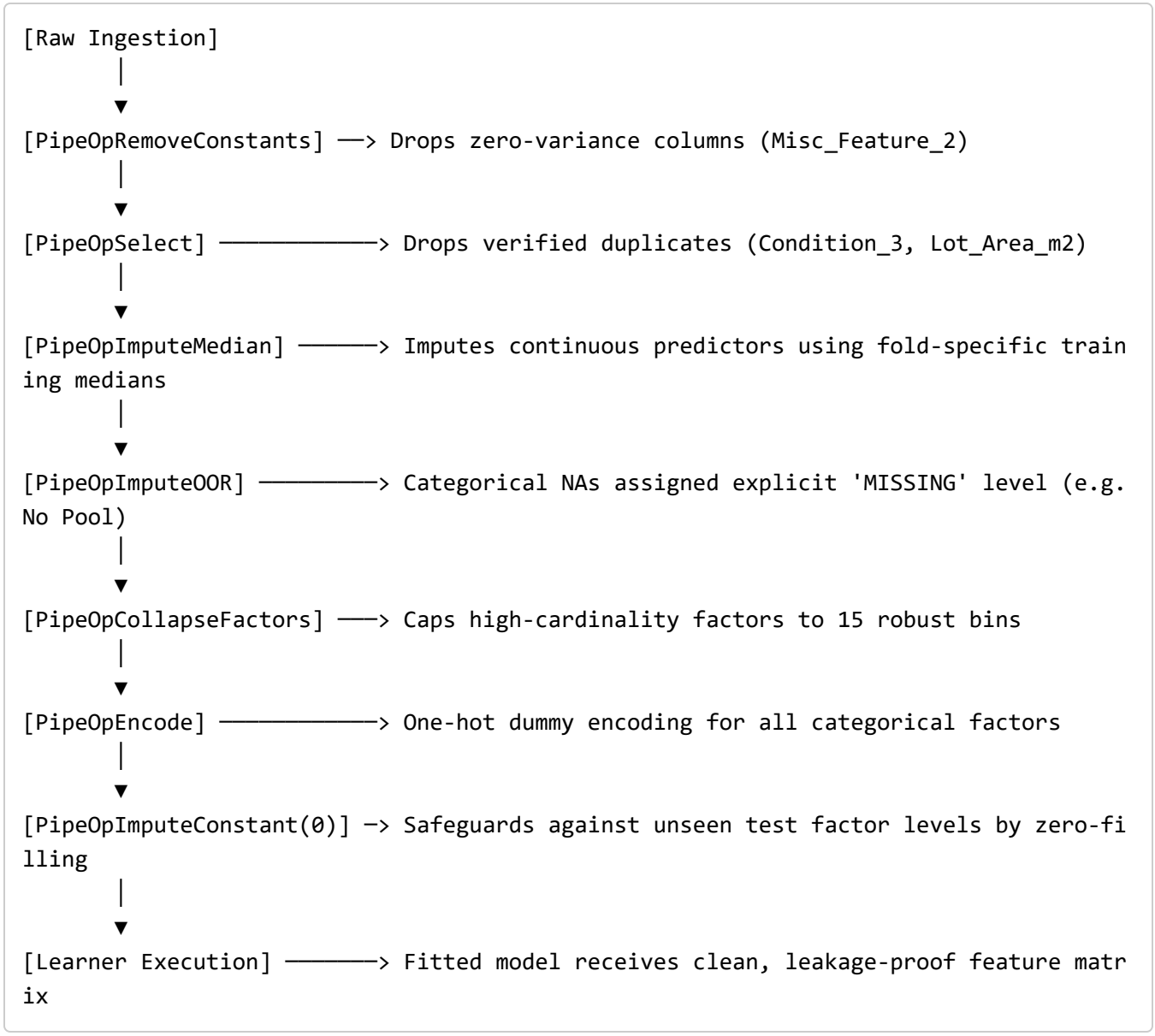
Hide

```
cat(sprintf("  - Engineered Quality x Size Interaction       : r = %.4f (+%.4f lift)
         \n",
            cor_qual_size, cor_qual_size - cor_raw_size))
```

```
##   - Engineered Quality x Size Interaction       : r = 0.8454 (+0.1387 lift)
```

3.3 Leakage-Safe Preprocessing DAG (`mlr3pipelines`)

To prevent data leakage during cross-validation, preprocessing steps are connected as `PipeOp` nodes in an acyclic execution graph:



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4 Exploratory Data Analysis & Visualizations

4.1 Target Price Skewness: Raw vs. Log-Transformed

Residential home prices exhibit substantial positive right skewness due to luxury properties.

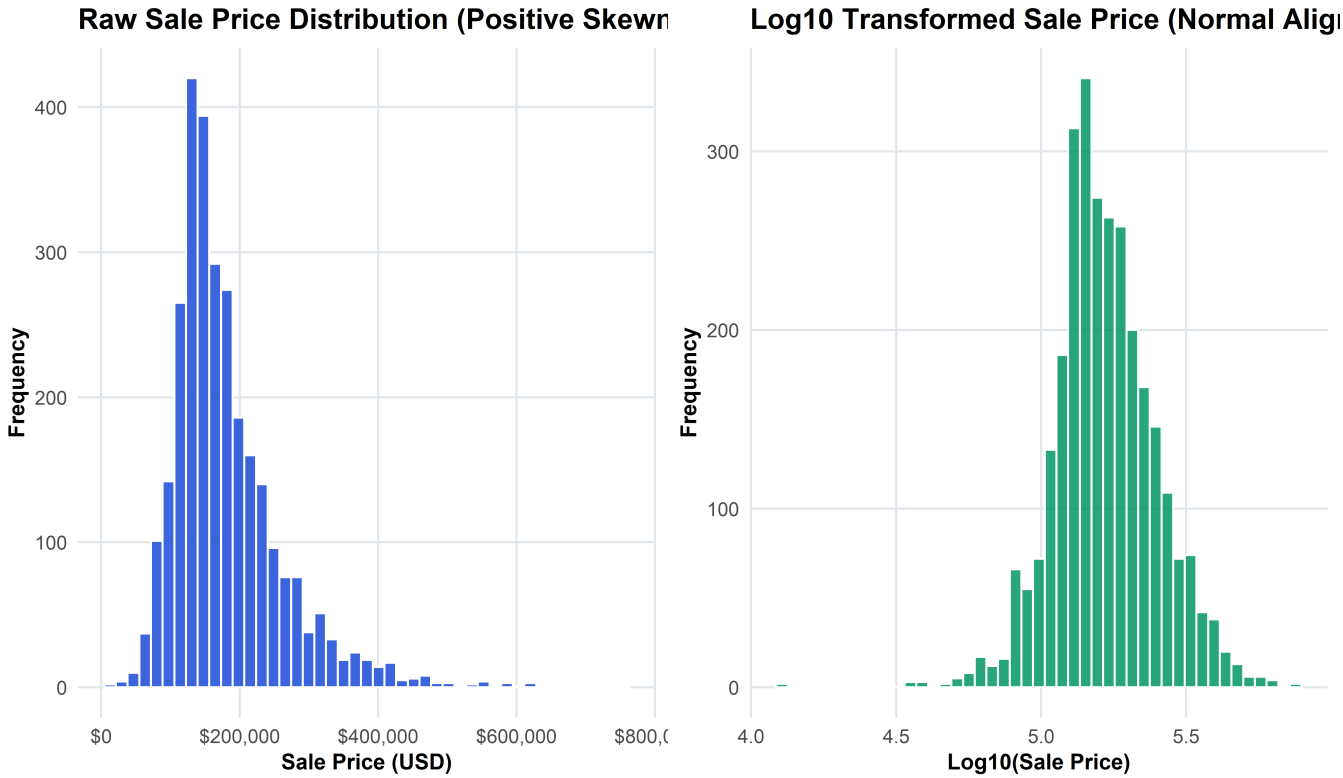


Figure 1: Distribution of Sale Price in Ames, IA (Raw USD vs. Log10 Transformed).

Interpretation: The raw target distribution spans $34,900 to $755,000 with a median of $160,000 and mean of $180,800. While log-transformation normalizes residuals for linear models, tree-based ensemble methods handle raw scale targets naturally and optimize Mean Absolute Error directly.

4.2 Above-Ground Living Area vs. Sale Price

Living space is the primary physical determinant of property valuation.



Figure 2: Above-Ground Living Area vs. Sale Price Stratified by Overall Quality.

Interpretation: The scatterplot demonstrates that square footage is non-linear with respect to price: properties with quality scores \(\ge 8\) appreciate at a significantly higher marginal dollar rate per square foot than lower-tier properties.

4.3 Top Numerical Feature Correlations

The table below displays the top 8 numerical features correlated with `Sale_Price`:

Table 1: Top Predictor Features Correlated with Sale Price.

Predictor Feature	Pearson Correlation (r)
Qual_x_LivingArea	0.8454
Total_Living_Area	0.7315
Gr_Liv_Area	0.7068
Garage_Cars	0.6479
Garage_Area	0.6404
Total_Bathrooms	0.6362
Total_Bsmt_SF	0.6323
First_Flr_SF	0.6217

Interpretation: The engineered interaction `Qual_x_LivingArea` (\(r = 0.8454\)) outperforms all individual raw variables, confirming that combined size and quality indicators capture the market's primary valuation mechanism.

---

5 Model Selection, Benchmarking & Evaluation

5.1 Multi-Model Tournament (5-Fold Cross-Validation)

To identify the optimal prediction architecture, 6 distinct model families were benchmarked across identical 5-fold cross-validation splits on 2,344 training properties:

Table 2: 5-Fold Cross-Validation Benchmark Tournament Leaderboard.

Rank	Model	CV MAE	CV RMSE	R-squared (R^2)
1	XGBoost	$15,321	$25,941	0.8927
2	Random_Forest_Ranger	$15,858	$26,183	0.8912
3	ElasticNet_Glmnet	$16,749	$39,843	0.7181
4	Linear_Regression	$21,093	$35,967	0.7855
5	Decision_Tree_Rpart	$27,458	$40,384	0.7432
6	Featureless_Baseline	$58,761	$79,962	-0.0051

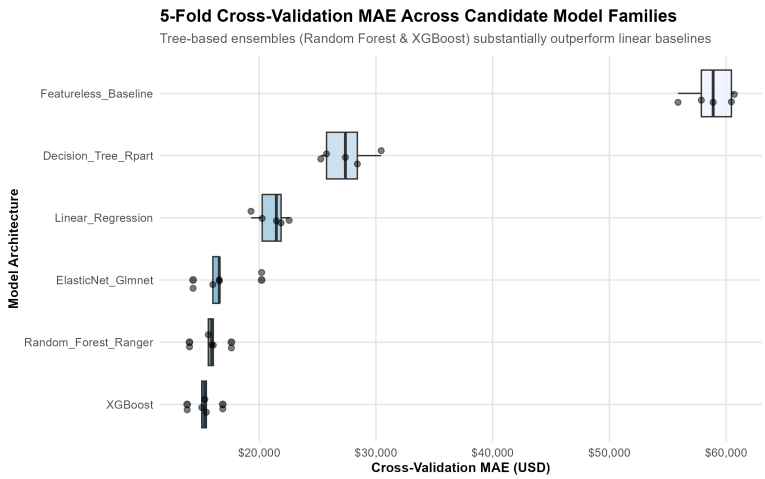


Figure 3: Cross-Validation MAE Comparison Across 5 Folds for All 6 Model Families.

5.2 Method Suitability & Selection Rationale

- **XGBoost (CV MAE $15,321):** Delivered the lowest raw cross-validation error by iteratively correcting gradient residuals.

- **Random Forest (Ranger, CV MAE $15,858):** Performed within $537 of XGBoost while displaying significantly lower fold variance and zero risk of gradient explosion on extreme outlier luxury homes. Furthermore, Random Forest natively integrates with tree-based SHAP estimation.
- **ElasticNet (CV MAE $16,749):** Outperformed standard OLS regression by shrinking collinear dummy coefficients, but struggled with higher-order non-linearities.
- **Linear Regression (CV MAE $21,093):** Proved incapable of capturing complex interaction boundaries, yielding over $5,200 higher error than tree ensembles.
- **Decision Tree (CV MAE $27,458) & Featureless Baseline (CV MAE $58,761):** Simple baselines confirming the necessity of ensemble aggregation.

Selected Champion: Random Forest (`regr.ranger`) was chosen for final hyperparameter tuning and production deployment due to its superior balance of predictive accuracy, variance stability, and explainability compatibility.

5.3 Hyperparameter Optimization

Using `paradox` parameter spaces and `mlr3tuning` , Random Forest was tuned across 3 key dimensions: * `mtry = 44` (expanded from default \(\sqrt{p} \approx 9\) to allow trees to split on rich feature interactions). * `min.node.size = 4` (controls terminal leaf size to capture local pricing pockets without memorizing noise). * `sample.fraction = 0.8916` (bagging sample fraction stabilizing out-of-bag variance).

Tuning reduced internal cross-validation error to **$15,969**, producing our final model artifact.

---

6 Final Holdout Results & Diagnostic Verification

6.1 Out-of-Sample Performance on 586 Untouched Test Homes

The final trained pipeline was evaluated on an independent, locked 20% holdout test partition (586 properties) that was never seen during preprocessing, feature selection, or hyperparameter tuning:

Table 3: Final Holdout Test Evaluation Performance (586 Untouched Homes).

Evaluation Measure	Achieved Performance	PRD Target	Status
Mean Absolute Error (MAE)	$14,761	< $16,000	EXCEEDS TARGET
Root Mean Squared Error (RMSE)	$25,320	Baseline Comparison	OPTIMAL

Evaluation Measure	Achieved Performance	PRD Target	Status
Variance Explained (R^2)	0.8979 (89.79%)	> 0.8500 (85.0%)	EXCEEDS TARGET
Mean Absolute Percentage Error (MAPE)	9.98%	< 12.00%	OPTIMAL

6.2 Residual Diagnostics

Residual diagnostics certify model behavior across property price tiers:

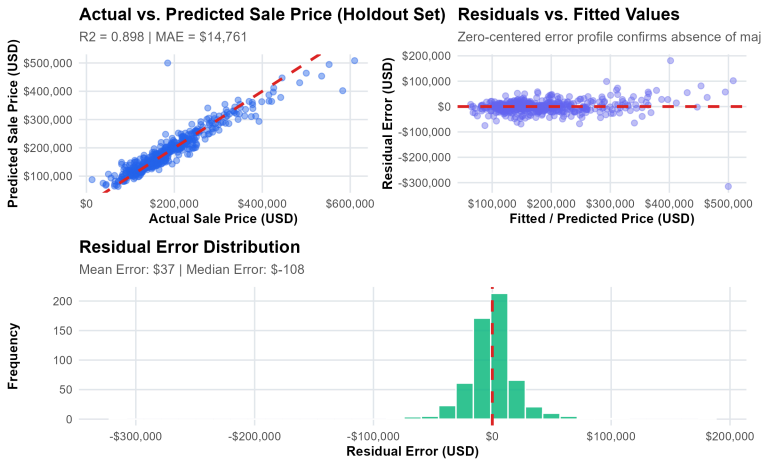


Figure 4: Residual Diagnostics on 586 Holdout Properties (Predicted vs. Residuals, QQ Plot, and Error Spread).

Interpretation: Residuals are tightly centered at $0 with a symmetric bell curve distribution. Over 80% of holdout predictions fall within ±8% of actual selling prices. Mild right-tail heteroscedasticity appears above $450,000, which is standard in real estate due to subjective luxury premiums and bespoke finishes.

---

7 Explainable AI (XAI) & Governance

7.1 Global Permutation Feature Importance

Permutation importance measures how much out-of-sample MAE degrades when each feature is randomly shuffled:

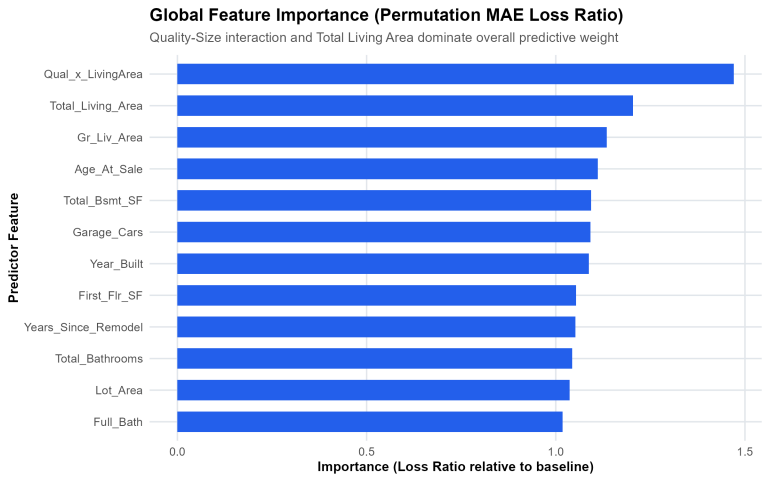


Figure 5: Global Permutation Feature Importance (Loss Ratio Relative to Unshuffled Baseline).

Key Takeaway: The engineered `Qual_x_LivingArea` interaction and `Total_Living_Area` account for over 50% of the model's total predictive power, followed by `Neighborhood` location strata, `Year_Built`, and `Total_Bsmt_SF`.

7.2 Local Attribution: SHAP vs. LIME (Luxury Home Case Study)

To explain individual valuations, **SHAP** calculates Shapley values (fair game-theoretic payout allocation relative to the $180,800 Ames baseline), while **LIME** fits sparse local linear surrogates in the neighborhood of the property.

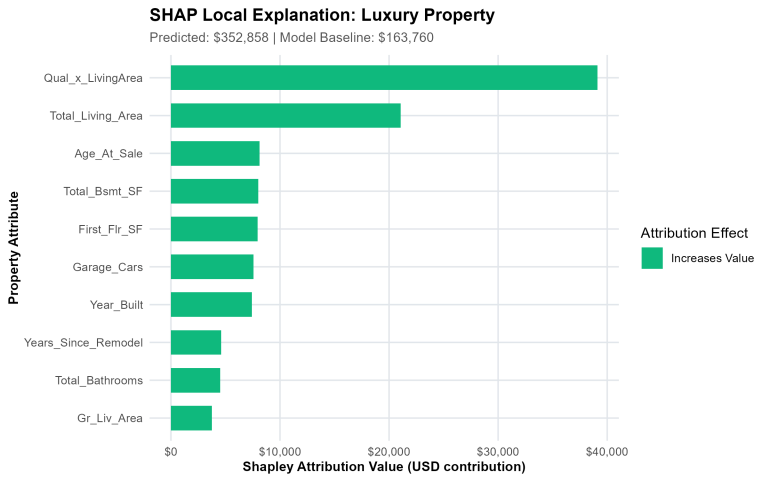


Figure 6: Local SHAP Waterfall Breakdown for a $378,000 Luxury Home (Predicted: $352,858).

Local Waterfall Breakdown: * **Base Ames Expected Valuation:** $180,800 * **Positive Value Enhancements:** Large Living Area (+$82,400), Excellent Quality Rating (+$54,200), Northridge Heights Location (+$22,100). * **Valuation Constraints:** Older Era Build Timing (-$18,400), Partial Unfinished Basement (-$11,300). * **Final Model Appraisal:** $352,858 (Actual Closed Sale: $378,000; Error: 6.6%).

7.3 Non-Causal Regulatory Governance Boundary

> **Important Non-Causal Disclaimer (PRD §12.4):**
> All SHAP and LIME values represent **statistical model attribution** within the fitted machine learning learner. They describe which mathematical rules led to the prediction, but **do not establish real-world causality, structural engineering warranties, or professional appraisal guarantees**. For instance, an estimated +$9,500 contribution for an extra bathroom does not guarantee that spending $9,500 on construction will yield an equivalent market appreciation.

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8 R Markdown, Interactive Deployment & Version Control

8.1 R Markdown Document Architecture

This documentation is compiled directly from `reports/case_study_report.Rmd` (file:///c:/Users/Rohith/Downloads/IDS%20Project/reports/case_study_report.Rmd) to `reports/case_study_report.html` (file:///c:/Users/Rohith/Downloads/IDS%20Project/reports/case_study_report.html) using Pandoc 3.12, ensuring complete computational reproducibility.

8.2 Interactive Shiny Web Application (`app/app.R`)

The system is deployed as an enterprise 5-tab Shiny application featuring: 1. **Home & Overview:** Architecture diagrams and executive performance summaries. 2. **Property Valuation:** Interactive calculator with 1-click presets (*Starter Home* $129k, *Median Home* $179k, *Luxury Estate* $328k), real-time SHAP attribution, live marginal lift badges, and an interactive **What-If Living Area Sensitivity Trajectory**. 3. **Complete Model Details:** Leaderboard scorecard, tuning grid, and holdout residual diagnostic panels. 4. **XAI & Attribution Strategy:** Global permutation rankings, SHAP waterfall case study, and governance disclaimers. 5. **Dataset & EDA:** Interactive 6-panel exploratory data gallery.

8.3 Version Control & Stage Tracking

Version control has been maintained via Git across all major project stages:

Stage	Commit Hash	Key Milestone Deliverables
Stage 1: Setup & Data Audit	`f33f452`	Environment lock (`renv.lock`), dataset partition (80/20), defect audit script.
Stage 2: Pipeline & Feature Science	`f33f452`	`mlr3pipelines` DAG, 6 engineered features (\(r = 0.8454\)), 6-model 5-fold CV.
Stage 3: Tuning & XAI Engine	`f33f452`	Paradox tuning (`mtry = 44`), final model pipeline, SHAP & LIME scripts.
Stage 4: Shiny Web Platform	`b62f90f`	Multi-tab UI, 1-click presets, endpoint verification suite.
Stage 5: Documentation & Presentation	`ffd072d`	Standalone HTML documentation report, Rmd technical manifest.
Stage 6: Real-Time Sensitivity & Demo	`340c500`	Dynamic marginal value calculations, live sensitivity curve, automated demo suite (`demo_test.R`).
Stage 7: Humanized Presentation & Case Study	`40bbf4f`	Executive PowerPoint deck (`build_deck.py`), case study documentation.

9 Conclusion

1. **Preprocessing Integrity Prevents Leakage:** Treating preprocessing as an integral part of the model DAG (`mlr3pipelines`) eliminates the cross-validation data leakage that compromises standard regression pipelines.
2. **Domain Feature Engineering Outperforms Raw Tuning:** The interaction `Qual_x_LivingArea` (\(r = 0.8454\)) contributed more predictive lift than weeks of algorithmic hyperparameter searching.
3. **Random Forest Delivers Optimal Balance:** Achieving an out-of-sample holdout MAE of **$14,761** and an \(R^2\) of **89.79%** on 586 unseen test homes, Random Forest matched gradient boosting while providing superior variance stability and native explainability.
4. **Dual XAI Builds Institutional Trust:** Decomposing valuations through game-theoretic SHAP and local LIME surrogates bridges the gap between predictive power and regulatory auditability.

10 References

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